Experiment 5

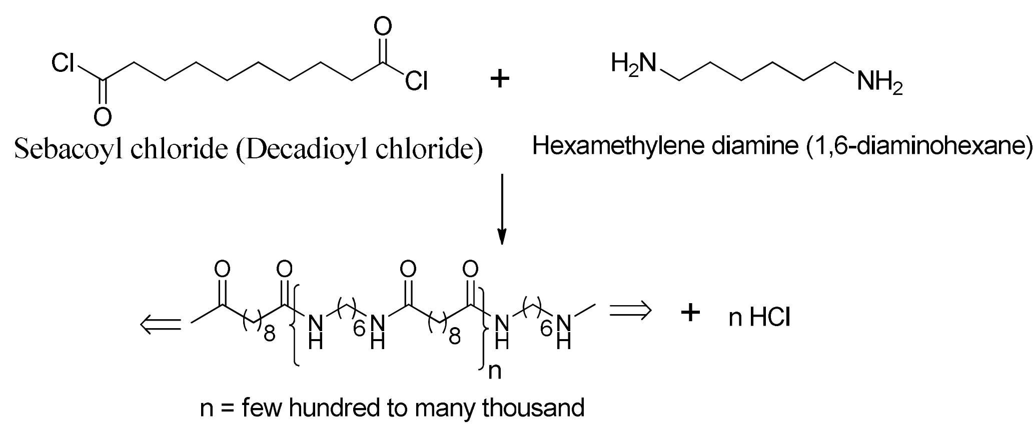
Nylon synthesis and chromatography

(This experiment to be done in groups of two)

Part 1: Synthesis of Nylon 6,10

Nylon was the first man-made fiber, discovered by Wallace Carothers at Du Pont in 1934. At the New York World's Fair in 1939, Du Pont advertised the introduction of Nylon stockings (hosiery) as a modern, improved, but less expensive substitute of silk stockings, which had been rather costly. Nylon stockings soon became the rage in fashion industry. The demand was far greater than the supply, so they were rationed until World War II, when nylon production was completely stopped. Nylon was then needed for parachutes and other wartime uses. At the end of the war, in 1945, it was a major news item that nylon again became available.

Nylon belongs to the class of polyamides, which are polymers where the monomer units are connected together by amide groups. In this experiment 1,6-hexanediamine or 1,6-diaminohexane (also commonly called hexamethylene diamine) is reacted with sebacoyl chloride (also called decadioyl chloride). Both amine ends of the 1,6-diaminohexane react with both acid chloride ends of the sebacoyl chloride to form new amide linkages. The polymer which forms is made up of alternating 1,6-diaminohexane and sebacoyl groups. This is called an alternating copolymer. This particular Nylon is made of units of 6 carbons between two nitrogen atoms, an amide linkage, and another unit of 10 carbons. This repeats many many times and therefore this is referred to as Nylon-6,10. Thus the first number (6) is represented by the diamine and the second number (10) is represented by the diacid chloride.



The reaction shown above belongs to the class of interfacial polymerization. Interfacial polymerization involves the reaction of monomer A dissolved in an organic phase with a monomer B dissolved in an inorganic phase at the interface between the two non-miscible phases. Interfacial polymerization can be of two types. In one case, there are two separate macroscopic layers in contact with each other. This is called unstirred interfacial polymerization. This is what we are going to study in this experiment. Unstirred interfacial polymerization is used to produce membranes and to create polymer by continuous removal in a single rope. The other case is one where one phase is dispersed as tiny droplets in the other (continuous) phase by using high speed stirring. This is called stirred interfacial polymerization. Stirred interfacial polymerization is used to produce tiny microcapsules (hollow inside) and microspheres (not hollow) for various applications like controlled release of drugs and pesticides.

PROCEDURE

1. Pour 20 mL of the solution of hexamethylene diamine in water made basic with sodium hydroxide in a 100 mL beaker. --- **Stock Solution A**
2. Gently pour 20 mL of **Stock Solution B**, which is sebacoyl chloride in hexane, using a glass rod into the 100 mL beaker in step 1. Be careful not to disturb the interface between the two solutions.
3. Two layers will be observed and nylon will be formed by the chemical reaction shown above at the interface where the two liquids meet. Avoid mixing the solutions by swirling the beaker.
4. Using a copper wire as a hook, hook the nylon at the center of the beaker and slowly pull it out as the 'nylon rope'. Use your glass rod as a spool to roll-out the continuously-made nylon. If you do it very carefully and slowly, you should be able to obtain an unbroken nylon strand until one of your reactants is used up. If your nylon strand snaps, use the wire to generate a new strand
5. After the entire polymer has been collected, which you will observe as a sudden break in the continuity of the nylon rope, wash it thoroughly with water and let it air dry.
6. Record the mass of nylon you have obtained after drying.
7. Observe the physical appearance of your product. Test the tensile strength of your nylon strand by stretching until it breaks.
8. Tape the formed nylon to your lab report.
9. Carefully dispose your chemicals into proper waste containers. **PLEASE NOTE THAT NO SOLUTION SHOULD BE DISPOSED DOWN THE DRAIN**.
10. Clean up your apparatus properly and wash your hands before you move on to the next experiment.
11. *Optional*: Food coloring dyes or phenolphthalein can be added to the lower (aqueous) phase to enhance the visibility of the liquid interface.

The following should be included in your report on the synthesis of Nylon:

1. Weight of the nylon formed
2. Qualitative tensile strength (e.g. high, low, medium)
3. Tape the formed nylon to your lab report.
4. Can you suggest an easy method to estimate the length of the polymer that you have prepared?

Part 2: Thin Layer Chromatogrtaphy (TLC)

Many of you have already seen chromatography if you have tried to remove a spot from some soiled clothes by applying a cleaning fluid. The stain simply diffuses outwards replacing the spot by a series of larger rings. In laboratories, chromatography is a powerful method used to separate and identify the components of a mixture. There are many different kinds of chromatography using different mobile phases or stationary phases.

Brief History: Chromatography was invented in 1906 by a Russian botanist named Mikhail Semenovich Tswett. His name, Tswett, in Russian, means 'color', which is an interesting coincidence, considering the nature of his most important discovery. Chromatography comes from the Greek words: chroma means 'color' and graphein means 'to write'. So chromatography means "color-writing" and a chromatograph means "written in color".

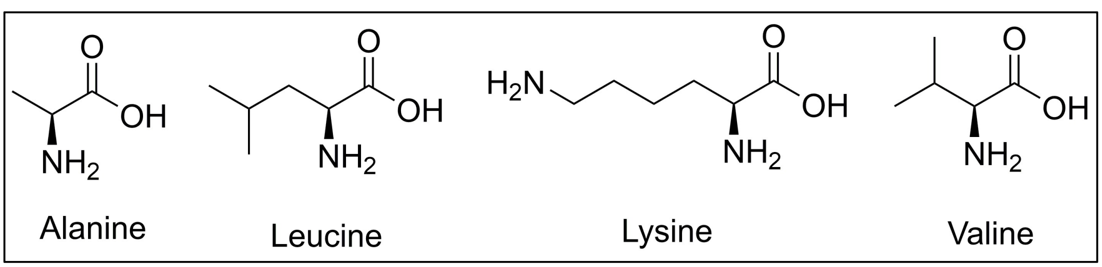
Chromatography separates compounds on the basis of:

1. Their solubility in the solvent used (the mobile phase) or
2. Their affinity (ability to hold tightly) for the stationary phase.

As the mobile phase travels across a layer of stationary phase by capillary action, it will carry with it components of the mixture. Those components that interact well with the mobile phase but poorly with the stationary phase will travel with the mobile phase; those that interact poorly with the mobile phase but strongly with the stationary phase will not travel as quickly. As the mobile phase moves, then, it carries components of the mixture at different rates. When the mobile phase is stopped, different components will have traveled different distances. Interactions between the components of the mixture and the stationary and mobile phases may include charge interactions or interactions between polar substances, including hydrogen bonding interactions. For instance, a TLC system can be set up with a very polar stationary phase and a nonpolar solvent. When a mixture containing substances of varying polarity is added to the mobile phase and carried across the stationary phase, the more polar compounds within the mixture will not travel as quickly. The less polar substances in the mixture will, over a limited amount of time, travel farther.

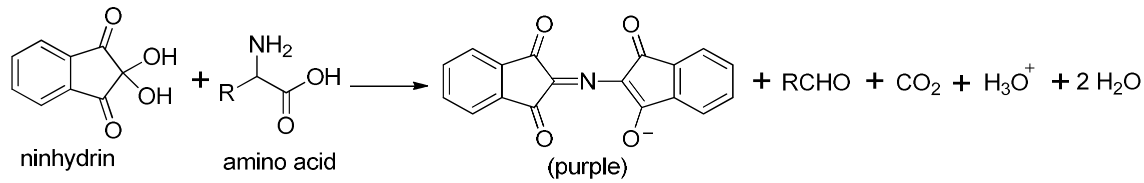
Thin layer chromatography (TLC) uses a thin layer of a solid stationary phase (such as porous silica gel) coated onto a glass, aluminum, or plastic support. Samples are placed on one end of the coated sheet. When the edge of the sheet is placed into a beaker of solvent, the solvent travels up the TLC sheet, like a wick, picking up and carrying components of the mixture with it. Our TLC sheets are coated with silica gel, which is very polar. *We will be using a mobile phase composed propanol and water* (**70:30, v/v**). The mobile phase is more nonpolar than the stationary phase. The system will be used to separate amino acids based on their degree of polarity.

Proteins, large molecules found in all living organisms, serve a variety of functions in metabolism, such as catalysis, transport, storage, control of growth and immune protection. Amino acids are the building blocks of proteins. Every amino acid has an amino group, a carboxyl group and a distinctive side-chain. Nature uses twenty different amino acids to synthesize proteins. The four amino acids that you will separate by thin layer chromatography are alanine, leucine, lysine, and valine. Their structures are shown below.



Structures of amino acids to be used today

Amino acids are colorless compounds. In order to see the spots on the chromatogram, you will apply a solution of ninhydrin to the paper. Ninhydrin will react with the amino acid to produce a purple compound.



Reaction between an amino acid and ninhydrin

PROCEDURE

1. You will be provided with different amino acid solutions, TLC plate and an unknown mixture of amino acids which you will have to separate on the TLC plate.
2. Lightly draw a line 1.0 cm above the bottom of the TLC plate with a pencil. This line will serve as a reference point to spot your samples to calculate your R_f values.
3. On the TLC plate, place a small spot of each of the four amino acids (alanine, leucine, lysine, and valine) and a spot of the unknown mixture with narrow capillary tubes provided (use a different capillary tube for each amino acid) along the line you have drawn. The spots should be very small and evenly spaced. If the spots are large, they will merge and/or have "tailing", that is, have a long tail.
4. Carefully place the amino acid plate into a developing chamber (250 mL beaker) that contains the mobile phase which is a mixture of isopropanol and water (**70:30, v/v**). Set it aside. The level of the solvent must be lower than the spots, so that the solvent can flow up by the capillary action past the spots.
5. Allow the solvent to rise to 3/4th of the way up the strip, remove it immediately, mark the solvent front and allow to solvents to evaporate off.
6. Dry the plate spotted with the amino acids on a hot plate in the hood. The hile amino acids are often invisible against the while silica gel of the TLC plate. Their position can be detected by spraying them lightly with a solution of ninhydrin (The solution contains 1.0 g of ninhydrin dissolved in 500 mL of n-butanol or n-propanol). Hold the TLC plate with a pair of tongs when spraying. Then, heat the TLC plate briefly on the hot plate in the hood to develop the color. CAUTION: *If you hold the plate with your fingers when spraying, your exposed skin will stain purple. The ninhydrin reacts with all amino acids including those in your skin. The purple stain will not wash off; it usually takes about two weeks to wear off as your skin is shed*. (Remember some of the organic compounds are toxic!)
7. Measure and record the distances traveled by the solvent front from the origin line on the chromatogram. Measure and record the distances traveled by the spots from the origin line.
8. Calculate the relation factor R_f values for the spots based on the following expression:

$$\boxed{R_f = \frac{\text{distance moved by compound from original spot}}{\text{distance moved by solvent from original spot}}}$$